Practical_2_Solutions

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1 Software Lab I - Practical 2

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4 Batch:B1

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5.1 Q1. Processed and Returned Orders

[1]:
```
processed_orders=[1152,1154,1155,1156,1157,1160,1161,1162,1166,1169,1170,1172,1176,1050,1178,1
returned_orders=[1153,1158,1159,1163,1164,1165,1167,1168,1171,1173,1174,1175,1177,1053,1055,10
orders=sorted(processed_orders+returned_orders)
print("a)",len(orders))
print("b)",orders[49])
print("c)","Processed" if orders[49] in processed_orders else "Returned")
print("d)",max(processed_orders))
print("e)",sorted(processed_orders)[:4])
```

```
a) 129
b) 1099
c) Processed
d) 1178
e) [1050, 1051, 1052, 1054]
```

5.2 Q2. Count vowels

[2]:
```
letters=['s','h','r','i','r','a','m','d','e','o','b','a','b','a','c','o','l','l','e','g','e']
vowels=['a','e','i','o','u']
count=0
for i in letters:
    if i in vowels:
        count+=1
print(count)
```

```
9
```

5.3 Q3. Repeated items in tuple

[3]:
```
t=(1,2,3,2,4,5,1,2,6,3,4,4)
for i in set(t):
    if t.count(i)>1:
        print(i,t.count(i))
```

```
1 2
2 3
3 2
4 3
```

5.4 Q4. Slice into two equal chunks and reverse

[4]:
```
l=[1,2,3,4,5,6,7,8]
mid=len(l)//2
a=l[:mid]
b=l[mid:]
print(a[::-1]+b[::-1])
```

```
[4, 3, 2, 1, 8, 7, 6, 5]
```

5.5 Q5. String letters

[5]:
```
s1="engineering"
s2="generation"
print("Only in first:",set(s1)-set(s2))
print("Common:",set(s1)&set(s2))
```

```
Only in first: set()
Common: {'n', 'e', 'i', 'g', 'r'}
```

5.6 Q6. Remove punctuations

[6]:
```
text="Hello!!! Python, is: awesome."
p='''!()-[]{};:'"\,<>./?@#$%^&*_~'''
result=""
for ch in text:
    if ch not in p:
        result+=ch
print(result)
```

```
Hello Python is awesome
```

6 Extra Practice

6.1 1. Products

[7]:
```
number=[8,9,9,1,6,9,5,7,3,9,7,3,4,8,3,5,8,4,8,7,5,7,3,6,1,2,7,4,7,7,8,4,3,4,2,2,2,7,3,5,6,1,1,
price=[195,225,150,150,90,60,75,255,270,225,135,195,30,15,210,105,15,30,180,60,165,60,45,225,1
print("a)",len(set(price)))
print("b)",sum(number))
avg=sum(price)/len(price)
print("c)",avg)
print("d)",max(price))
rev=0
for q,p in zip(number,price):
    rev+=q*p
print("e)",rev)
print("f)",number[19]>number[49])
exp=0
for p in price:
    if p>avg:
        exp+=1
print("g)",exp)
```

```
a) 20
b) 490
c) 139.01960784313727
d) 285
e) 67100
f) True
g) 52
```

6.2 2. Top three students

[8]:
```
students=((1,"A","Nagpur",80,90,95),(2,"B","Pune",75,88,91),(3,"C","Delhi",98,92,95),(4,"D","M
tot=[]
for s in students:
    tot.append((s[1],sum(s[3:])))
tot.sort(key=lambda x:x[1],reverse=True)
print(tot[:3])
```

```
[('C', 285), ('G', 276), ('E', 273)]
```

6.3 3. Common letters

[9]:
```
a="python"
b="programming"
print(set(a)&set(b))
```

```
{'p', 'o', 'n'}
```

6.4 4. List rotation

[10]:
```
L=[1,2,3,4,5]
k=3
for i in range(k):
    L=L[1:]+L[:1]
    print(L)
```

```
[2, 3, 4, 5, 1]
[3, 4, 5, 1, 2]
[4, 5, 1, 2, 3]
```

6.5 5. Sum of third elements

[11]:
```
t=[(1,2,3,4,5),(5,6,7,8,9),(9,8,6,5,4)]
s=0
for i in t:
    s+=i[2]
print(s)
```

```
16
```

6.6 6. Word frequencies

[12]:
```
text="python is easy python is powerful and python is fun"
words=text.split()
freq={}
for w in words:
    freq[w]=freq.get(w,0)+1
print(freq)
```

```
{'python': 3, 'is': 3, 'easy': 1, 'powerful': 1, 'and': 1, 'fun': 1}
```